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Disease control/Moyens de lutte

Evaluation of disease management approaches for powdery mildew on *Cannabis sativa* L. (marijuana) plants

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Abstract: Powdery mildew on cannabis (*Cannabis sativa* L., marijuana), caused by *Golovinomyces cichoracearum*, reduces plant growth and overall quality. To investigate disease management options, biological, chemical and physical approaches were assessed. A mildew-susceptible strain, ‘Copenhagen Kush’, was grown indoors with continual exposure to mildew inoculum. Treatments were applied weekly over a four-week period to groups of four plants once mildew infection had established itself. Trials were repeated thrice under varying initial disease pressures. Disease assessments were made weekly and the percentage of area infected on 30 leaflets per plant was used to calculate a disease rating score for treated and control plants. Disease progress curves were plotted and AUDPC values were determined for each treatment. To test the effect of UV-C light on mildew development, plants were exposed daily for 3–5 s over 28 days to UV-C light. The response of 12 cannabis strains to powdery mildew infection was assessed after exposing them to inoculum over a period of two weeks. The most effective treatments that significantly ($P < 0.05$) reduced disease in three trials were Luna Privilege SC (fluopyram), Regalia® Maxx, MilStop®, Rhapsody ASO™, neem oil, and Stargus®. Treatments that were less effective included ZeroTol®, boric acid, and Actinovate® SP. Daily exposure of plants to UV-C light significantly reduced disease (by 45.2%, $P < 0.05$). Seven of 12 cannabis strains had significantly lower disease severity compared with the other five strains. The disease management strategies evaluated in this study have potential for reducing powdery mildew development on cannabis.

Keywords: biocontrol agents, disease control, marijuana, powdery mildew, UV-C

Résumé: Sur le cannabis (*Cannabis sativa* L., marijuana), l'oïdium, causé par *Golovinomyces cichoracearum*, réduit la croissance et altère la qualité des plants. Afin d'étudier la gestion de la maladie, des approches biologiques, chimiques et physiques ont été évaluées. Une souche sensible à l'oïdium, ‘Copenhagen Kush’, a été cultivée à l'intérieur et exposée continuellement à l'inoculum de l'oïdium. Des traitements ont été appliqués hebdomadairement pendant quatre semaines à des groupes de quatre plants à partir du moment où l'infection est survenue. Les essais ont été répétés trois fois sous différentes pressions initiales de la maladie. Cette dernière a été évaluée toutes les semaines et le pourcentage de surface foliaire infectée de 30 folioles par plant a été utilisé pour calculer une cote de classement de la maladie pour les plants traités et les plans témoins. Des courbes d'évolution de la maladie ont été tracées et les valeurs de l'AUDPC ont été déterminées pour chaque traitement. Pour tester l'effet des rayons UV-C sur l'évolution de l'oïdium, les plants y ont été exposés quotidiennement pendant 3 à 5 s pendant 28 jours. La réaction de 12 souches de cannabis à l'oïdium a été évaluée après les avoir exposées à l'inoculum pendant deux semaines. Les traitements qui ont le plus efficacement ($P > 0,05$) réduit l'incidence de la maladie au cours de trois essais étaient Luna Privilege SC (fluopyrame), Regalia Maxx, MilStop, Rhapsody ASO, l'huile de margousier et Stargus. Les traitements qui étaient moins efficaces incluaient ZeroTol, l'acide borique et Actinovate SP. L'exposition quotidienne des plants aux rayons UV-C a également réduit significativement l'incidence de la maladie (de 45,2%, $P < 0,05$). Sept des douze souches de cannabis affichaient une plus faible gravité de la maladie comparativement aux cinq autres souches. Les stratégies de gestion de la maladie évaluées au cours de cette étude peuvent réduire l'évolution de l'oïdium sur le cannabis.

Mots clés: agents de lutte biologique, lutte contre les maladies, marijuana, oïdium, rayons UV-C

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Introduction

Powdery mildew diseases affect a wide range of host species and are caused by a large number of different obligate parasites belonging to several genera (Agrios 2005). On cannabis (*Cannabis sativa* L., marijuana), powdery mildew in Canada has been reported to be caused by *Golovinomyces cichoracearum sensu lato* (Pépin et al. 2018), and by closely related *Golovinomyces* species on hemp in other parts of North America (Schoener and Wang 2018; Gwinn et al. 2019; Ocamb and Pscheidt 2019; Szarka et al. 2019; Weldon et al. 2020). Powdery mildew is a prevalent disease that affects plant production in indoor controlled environments, greenhouses, and outdoor locations (Punja 2018; Punja et al. 2019). Following spore germination and infection, which can occur on leaf and stem surfaces, colonies that develop can limit photosynthesis and reduce nutrient availability to the plant, causing premature leaf drop as well as reducing overall vigour and potential yield of plants. Powdery mildew diseases, in general, have been managed by applications of fungicides and other chemical products, biological control agents, induction of disease resistance, and development of cultivars with genetic resistance (Agrios 2005). On cannabis, powdery mildew disease management relies on cultural control methods, such as removal of diseased leaves (deleafing), maintaining relative humidity at levels not conducive for pathogen development, applications of vaporized sulphur or reduced risk chemicals such as potassium bicarbonate (MilStop®) or hydrogen peroxide (ZeroTol®) (Health Canada 2019). There are currently no synthetic chemical fungicides registered for powdery mildew control on cannabis in Canada unlike on other crops. While reduced risk chemical products including potassium bicarbonate, hydrogen peroxide and lactic acid are registered for use on cannabis in Canada (Health Canada 2019), data on their comparative efficacy is lacking. In addition, the assessment of other potential methods for disease control, including biological control agents, has not been previously conducted. The rapid expansion of the cannabis industry in Canada requires that adequate disease control options for powdery mildew be identified for producers through an assessment of available options that do not impose additional risk to consumers and that meet Health Canada's requirements for use (Health Canada 2019). Since strains (genotypes) of cannabis can also vary in their susceptibility to powdery mildew infection, an assessment of the levels of resistance to this disease requires further study.

In this study, the efficacy of nine reduced risk chemical or biological control products in managing powdery mildew

development on cannabis plants was evaluated (Table 1). These products included three biological control agents, namely Actinovate® SP (containing *Streptomyces lydicus* strain WYEC 108), Rhapsody ASO™ (containing *Bacillus subtilis* strain QST 713), and Stargus® (containing *Bacillus amyloliquefaciens* strain F727). Chemical treatments evaluated included MilStop (potassium bicarbonate), ZeroTol (hydrogen peroxide), Silamol® (orthosilicic acid), neem oil, boric acid, and the plant defence inducer Regalia® Maxx (containing an extract of *Reynoutria sachalinensis*). The fungicide Luna® Privilege SC 500 (fluopyram) was included as an industry standard for comparative purposes. Luna Privilege Greenhouse, which makes use of the same active compound, but at a higher concentration, is registered for use in Canada for the control of powdery mildew and botrytis diseases on greenhouse grown cucumbers, peppers, tomatoes as well as other diseases (Bayer 2019). In addition, the efficacy of UV-C light to manage powdery mildew was also examined by daily exposure of leaves. Finally, the susceptibility of 12 strains (genotypes) of cannabis to disease development was assessed to determine their resistance to powdery mildew.

Materials and methods

Plant materials and inoculation

Vegetative cuttings (approximately 12 cm in height) of the powdery mildew susceptible strain 'Copenhagen Kush' (CPH) were obtained from stock (mother) plants maintained in an indoor growing room with 24-hr lighting and rooted in a mixture (3:1) of coco (Canna Coco) and perlite (Dutch Treat) in individual 8.5 cm² pots. The pots were placed under humidity domes (18-cm height dome with vents) for 10–14 days at room temperature (22–25 °C) in an indoor growing room with ambient lighting. To maintain high humidity during this time, the insides of the domes were misted with water every other day. After this time, the domes were removed, and plants were placed under two Sunblaster brand (Langley, British Columbia, Canada) 54 watt 6400 k T5HO lights with a 24-h photoperiod. Plants were watered as needed with a solution of 1 mL/L Sensi Grow Coco pH Perfect A + B (Advanced Nutrients, West Hollywood, California, USA) and 1 mL/L General Hydroponics Calimagic (General Hydroponics, Santa Rosa, California, USA) adjusted to a pH of 5.8–6.2 using Advanced Nutrients pH-Down (Advanced Nutrients, West Hollywood, California, USA). To initiate powdery mildew, initial inoculum originating from cannabis plants grown in a licenced facility in British

Table 1. Products evaluated for management of powdery mildew on cannabis plants and the average percent disease reduction values obtained over three trials compared to untreated control plants.

Product	Active Ingredient	Source	Rate Used	Average disease reduction (%)		
				Trial 1	Trial 2	Trial 3
Actinovate® SP	<i>Streptomyces lydicus</i> WYEC 108 0.037% (w/w)	Novozymes BioAg. Saskatoon, Canada	0.38 g/L	33.26	6.73	20.56
Boric acid	2% boron	Sigma-Aldrich. Missouri, USA	12.5 mL/L	22.23	45.32	37.7
Luna Privilege SC 500	40.98% fluopyram	Bayer CropScience Inc. New Jersey, USA	5 mL/L	78.26	79.09	76.92
MilStop®	85% potassium bicarbonate	BioWorks Inc. New York, USA	2.4 g/L	82.72	76.55	72.07
Neem oil	70% cold pressed neem oil	Terramera, Inc. British Columbia, Canada	1 mL/L	80.6	59.53	70.55
Regalia Maxx	20% extract of <i>Reynoutria sachalinensis</i>	Marrone Bio Innovations. California, USA	2.5 mL/L	80.25	32.24	71.83
Rhapsody ASO™	QST 713 dried <i>Bacillus subtilis</i> , contains a minimum of 1×10^9 CFU/g	Bayer CropScience Inc. New Jersey, USA	15 mL/L	55.61	35.43	71.83
Silamol®	2.5 % available silicon [Si(OH)4]	Frontline Growing Products Ontario Canada	2.4 mL/L	24.77	73.19	69.26
Stargus®	96.4% <i>Bacillus amyloliquefaciens</i> strain F727 cells and spent fermentation media. Contains a minimum of 1×10^9 CFU/g	Marrone Bio Innovations. California, USA	15 mL/L	52.4	41.41	63.7
ZeroTol®	27% hydrogen peroxide	BioSafe Systems. Connecticut, USA	10 mL/L	51.99	16.92	38.27

All products were applied as foliar sprays to plants until run-off. Applications were made weekly for four weeks. Disease was rated weekly starting before the first application. Average disease reductions were calculated from the differences in the AUDPC values between the control and the treatment specific to that trial.

Columbia was transferred to healthy plants by collecting spores and dusting plants. A single heavily infected plant was then placed in the centre of the room and allowed to release conidia. This ensured that background levels of inoculum in the indoor room where the experiments were to be conducted were sufficient for the disease to establish naturally. Rooted plants were removed from the high humidity environment, placed on a table, and received one spray of MilStop to minimize any variance in the initial levels of infection. The plants were then maintained for seven days in the growing room before treatments were applied (Table 1).

Treatment applications and disease assessments

The plants were arranged in groups of four per treatment, unless otherwise noted. In each experiment, two or three different treatments were compared to a control group. Products to be tested were prepared on the day of application and sprayed onto plants using a pressurized hand held spray bottle to runoff. At the start of the experiment (week 0), plants were assessed for powdery mildew infection, after which they received the appropriate treatment. Disease was assessed on the 30 most diseased leaflets (assessed visually) per group by placing them into one of five categories, based on the percent coverage of mildew (1–20%, 21–40%, 41–60%, 61–80%, or 81–100%). To calculate a disease rating score, the

number of leaflets in each category was multiplied by a factor of 0.1, 0.3, 0.5, 0.7, or 0.9, respectively (with these numbers representing the midpoint of each category). Plants received four weekly sprays in total and five disease assessments were made over the span of five weeks, each at weekly intervals. Luna Privilege SC 500 (Bayer CropScience, Research Triangle Park, North Carolina, USA) was only applied once, at the beginning of the experiment. Each experiment was repeated three times for each treatment (referred to as trials).

Data analysis

The average disease rating score for each treatment per week was plotted over the duration of the experiment to obtain a disease progress curve for each treatment in each of the trials. The results from each trial are presented separately as there were differences in the initial levels of infection on the plants as well as the overall disease pressure during the trial (see Table 2). To statistically show these differences between trials, the disease ratings were compared by ANOVA followed *post hoc* by Tukey's HSD test at $P < 0.05$ and are presented in Table 2. To test for significant differences between treatments within each trial, these data were also subjected to ANOVA followed *post hoc* by Tukey's HSD test at $P < 0.05$ and are presented in Table 2. Finally, area under the disease progress curve (AUDPC) was calculated from the disease progress curves (Simko and Piepho 2012) and compared

Table 2. Comparisons of final (week 5) disease rating scores in response to different treatments to manage powdery mildew on cannabis plants. Comparisons were made between trials and between treatments within a trial.

Treatment	Trial 1	Trial 2	Trial 3
Regalia [®]	1.15 a A	19.95 b A	5.15 a A
Rhapsody TM	9.7 a C	16.6 a A	9.5 a A
MilStop [®]	3.2 a AC	3.2 a B	5.1 a A
Control 1	25.28 a D	26.8 a C	27 a B
ZeroTol [®]	7.18 a A	19.8 b A	12.25 ab A
Actinovate [®]	13.55 a A	22.67 b A	21.55 b B
Control 2	22.15 a B	26.1 ab A	26.85 b C
Boric acid	21.4 a A	18.55 a A	20.1 a A
Silamol [®]	16.95 a A	3.28 b B	10.08 c B
Control 3	27 a B	26.35 b C	27 a C
Stargus [®]	7.08 a A	12.55 b A	4.13 a A
Neem	1.95 a B	8.1 b B	3.65 ab A
Luna	1.75 ab B	3.7 a C	0.78 b A
Control 4	22.05 a C	25 a D	20.8 b B
CleanLight TM	11.74 a A	3.68 b A	–
Control 5	23.9 a B	26.34 b B	–

The average disease severity scores for each treatment per trial are shown at week 5. Significant differences between trials of the same treatment are indicated by a lowercase letter, while uppercase letters signify significant differences between different treatments in the same trial. There were four replicate plants per trial except in the case of the CleanLight trials ($n = 8$) and the second trial of ZeroTol, Actinovate and their respective control ($n = 3$). No comparisons are made between treatments that are not grouped together. Different letters represent significant ($P < 0.05$) differences, as determined through ANOVA with Tukey's *post hoc* test.

for differences among treatments within each trial by ANOVA followed *post hoc* by Tukey's HSD test at $P < 0.05$. These results are presented in Figs 3–6. *UV treatments for powdery mildew control*

Plants of strain CPH were grown as previously described and arranged in groups of 8, with 8 untreated (control) plants per trial. The treated plants received daily exposure to UV-C light from a hand-held CleanLightTM Pro (Honselersdijk, NL) unit over a period of four weeks. Plants were treated for 3–5 s per day (equal to 3–6 mJ/cm² of radiation, as per the manufacturer) by moving the unit uniformly over and around the plant at a distance of 5 cm away. Treated and control plants were rated for disease as previously described, with the 30 most diseased leaflets being assessed at weekly intervals five times over the duration of the trial. Disease rating scores were calculated for treated and control plants each week and graphed, and AUDPC values were calculated as previously described. This experiment was conducted two times. Data are presented separately for each trial (Fig. 7). The disease rating scores and AUDPC values were compared for significant differences between treated

and control plants using ANOVA followed *post hoc* by Tukey's test at $P < 0.05$ (Fig. 7).

Scanning electron microscopy

To compare the extent of powdery mildew development (mycelial growth and sporulation) between control plants and those receiving specific treatments, leaf samples ca. 0.5 cm² in size were prepared for scanning electron microscopy as described by Punja (2018). The leaves were obtained from untreated (control) plants with young and old powdery mildew colonies, as well as from plants receiving four weekly treatments of Regalia, Rhapsody, or MilStop. In addition, leaves from plants exposed to CleanLight (daily exposure for 28 days) were included. The extent of mycelial growth of powdery mildew and the presence of sporulation were visually compared for control and treated leaves.

Effect of Regalia Maxx on chlorophyll 'a' and 'b' levels

On plants receiving four applications of Regalia Maxx, it was observed that leaves appeared darker green when compared to control plants. Therefore, methanol extraction and spectrophotometric analysis of chlorophyll *a* and *b* in leaf samples was performed according to Sumanta et al. (2014). Two trials were conducted – one in an indoor growing room and the other under greenhouse conditions. Leaf samples were collected from approximately mid-height after the plants had received 4 applications, with 3–5 replicate samples. Mean values for chlorophyll *a* and *b* from the Regalia Maxx treated plants and the untreated control plants were compared for significant differences using ANOVA followed *post hoc* by Tukey's test at $P < 0.05$. Data for the two trials are presented separately.

Screening of strains for susceptibility to powdery mildew

Stock plants of 12 cannabis strains grown in an indoor room were used as a source of cuttings. The plants received two weekly sprays of MilStop (BioWorks, Victory, New York, USA) to reduce powdery mildew infection, after which cuttings were taken and rooted as described previously. The cuttings were dusted with powdery mildew spores from fresh infected leaf tissue before being placed under the humidity dome. Two visibly infected plants of strain CPH were also placed under the domes to provide an additional source of inoculum. In each trial, there were two groups of four plants arranged around two diseased plants. Cuttings

were left under the domes for a period of two weeks, during which the inside of the domes was misted on alternate days to maintain high humidity. At the end of two weeks, individual plants were rated for disease by placing them into one of five categories based on the percentage coverage of powdery mildew on the leaves, as described previously. The average disease score per strain was determined and graphed. Significant differences between the disease scores were determined using ANOVA followed *post hoc* by Tukey's test at $P < 0.05$. The experiment was conducted twice.

Results

Development of the powdery mildew fungus

The symptoms of powdery mildew infection on cannabis leaves were quite apparent as white mycelial growth and sporulation of the pathogen on the upper surface of leaves on plants in the growing room (*Fig. 1a*). Under the scanning electron microscope, a cross-section through a leaf showed mycelial growth on the epidermal surface (*Fig. 1b*), and

sporulation was abundant on the leaf surface (*Fig. 1c*). The production of spores in chains is a characteristic of the genus *Golovinomyces* (Pépin et al. 2018).

Treatment applications and disease assessments

The foliar treatments applied to manage powdery mildew had different levels of efficacy when compared to the untreated controls. The results from each of the three trials conducted for each product showed some variation between trials, which was attributed to differing levels of initial infection and/or disease pressure during the trials. The disease progress curves for each trial for each product tested are shown in *Fig. 2*. They show the varying rates and levels of disease development over time. A comparison of final, week 5, disease rating scores in response to different treatment applications to manage powdery mildew on cannabis plants is presented in *Table 2*. Statistical comparisons were made between trials and between treatments within a trial. The statistical analysis showed that disease levels on control plants in each of the experiments, i.e., Control 1 to Control 5 values, were in the range of 20.8 to 27, and thus showed

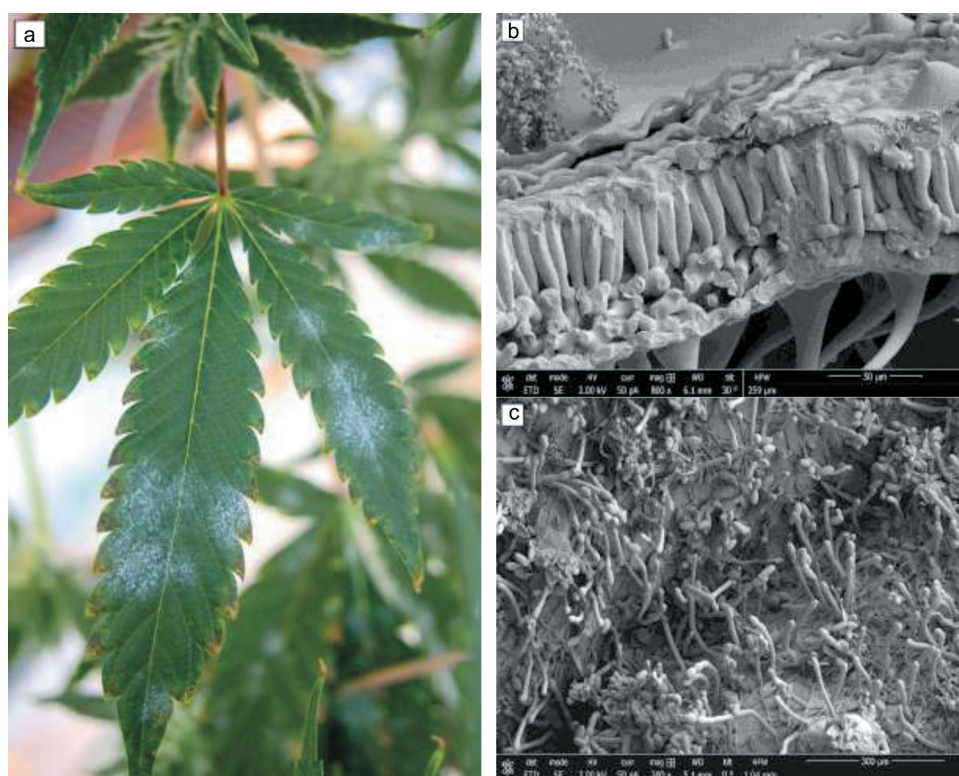


Fig. 1 Development of powdery mildew on cannabis leaves. (a) Initial infections are visible as white colonies on the upper surface of leaves. (b) Scanning electron microscopic image through a cross section of a diseased leaf showing powdery mildew mycelium growing over the surface of epidermal cells. The underlying cells of the epidermis and mesophyll layer, and the lower epidermis, can be seen in this section. (c) Abundant spore production from an older powdery mildew colony on the leaf surface.

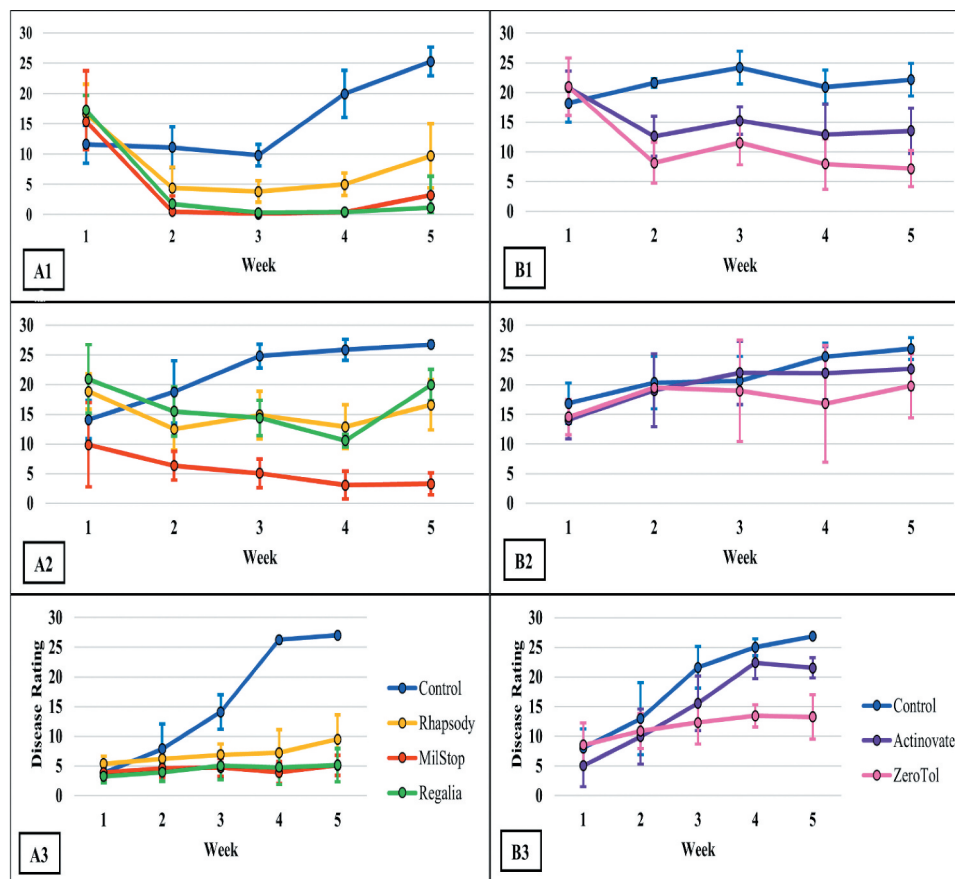


Fig. 2 The disease progress curves for powdery mildew development on cannabis plants when treated with Regalia Maxx, Rhapsody ASO, MilStop® (A1-3), ZeroTol®, Actinovate® (B1-3), boric acid, Silamol® (C1-3), Stargus®, neem oil, Luna Privilege SC 500 (D1-3), CleanLight™ UV-C (E1-2), and their respective untreated controls. Data represent the average disease score for each treatment in each week. There were four replicate plants per trial except in the case of the CleanLight trials ($n = 8$) and the second trial of ZeroTol, Actinovate and their respective control ($n = 3$). Error bars are 95% confidence intervals (continued on next page).

fairly consistent final disease levels in all trials. However, in a number of treatments, the data from different trials were significantly different. For example, in trial 2, the values for disease ratings were significantly different for Regalia, ZeroTol, Actinovate, Stargus, and neem when compared to the values for trials 1 and 3 (Table 2). The week five disease rating scores, however, showed that most treatments significantly reduced the development of powdery mildew in all three trials, except for ZeroTol, boric acid and Actinovate (Table 2), which were the poorest performing products.

Product efficacy based on area under the disease progress curve

The AUDPC values were calculated from the disease progress curves and are presented in Figs 3–7. The most effective products, which provided significant ($P < 0.05$) reductions in AUDPC values consistently in all three trials were: Luna Privilege SC 500; MilStop; neem oil; Regalia Maxx; Rhapsody ASO; and Stargus. These products also provided average disease reductions of over 50% (Table 2) for the majority of the trials. Products showing significantly reduced AUDPC values in two out of three trials were ZeroTol, boric acid, and

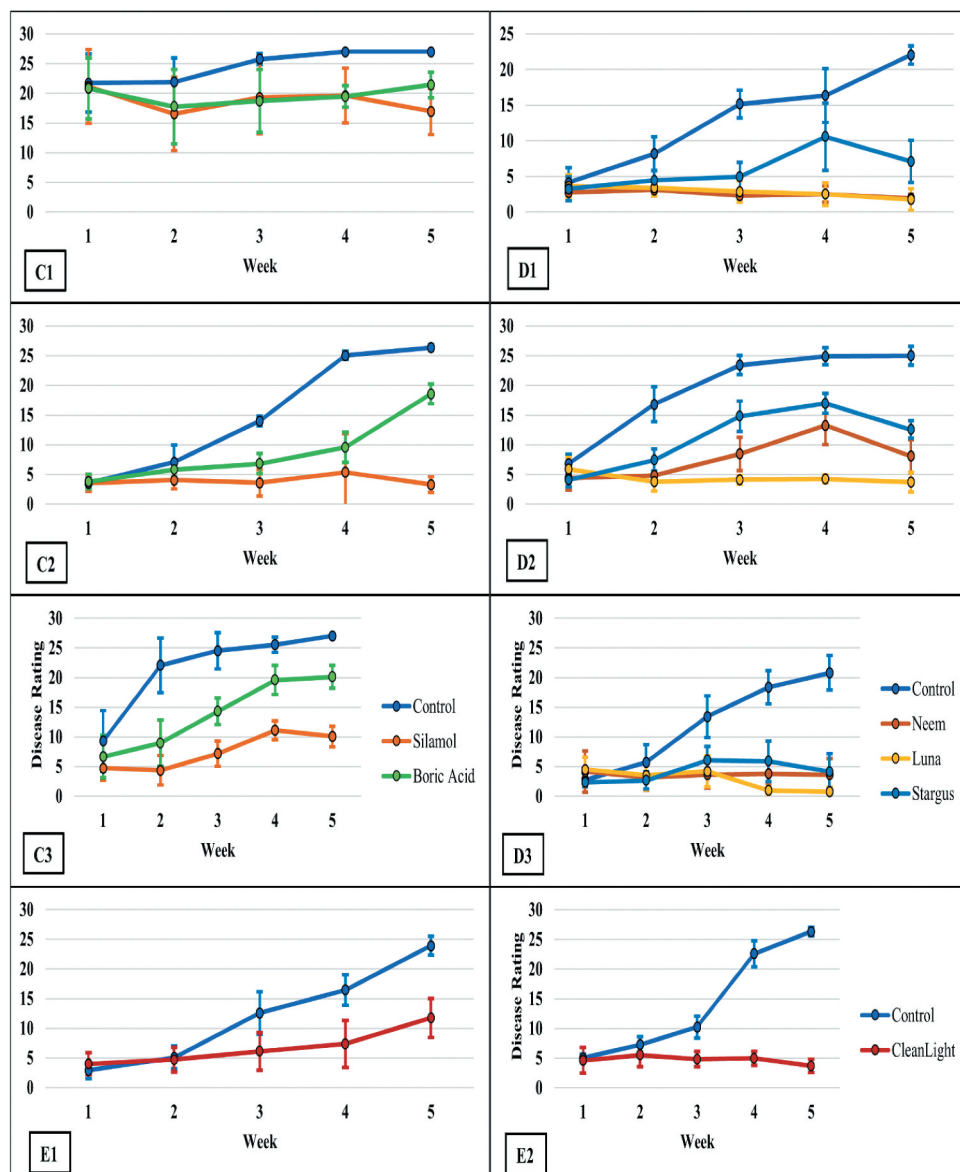


Fig. 2 (Continued).

Silamol, although these products often reduced disease by less than 50% in most of the trials (Table 2), with the exception of Silamol. Actinovate only provided significant disease control in one of the three trials. Plants receiving daily exposure to CleanLight had significantly ($P < 0.05$) lower AUDPC values in both trials, when compared to their appropriate controls (Fig. 7). The AUDPC values and the week five disease rating scores, therefore, provided similar conclusions on the efficacy of the products tested. The percentage reduction in disease levels for each of the products in each trial based on AUDPC values is summarized in Table 2. These values

were derived by comparison of AUDPC values for the treatments with the untreated control used in each trial.

Visual representation of product efficacy

Representative leaves were sampled from plants in each of the treatments and photographed to show the differences in product efficacy (Fig. 8). While the leaves were from different experiments, they were selected to represent the disease rating values shown in Table 2. When compared to the untreated control leaves, treatments with CleanLight, Luna, Regalia Maxx and MilStop showed the lowest

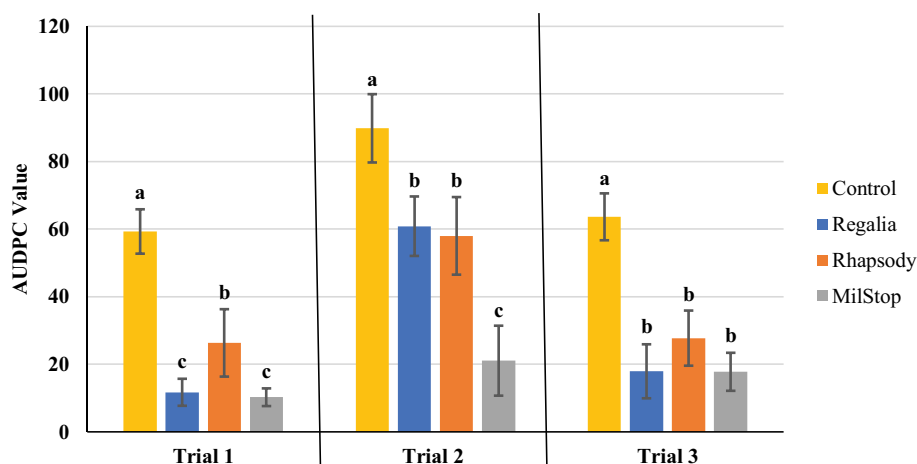


Fig. 3 Effect of four weekly applications of Regalia Maxx, Rhapsody ASO and MilStop® on development of powdery mildew on cannabis plants compared to a nontreated control. Data represent areas under the disease progress curves (AUDPC) from three repeated trials, each with four replicate plants. Error bars are 95% confidence intervals. Letters above the error bars represent significant differences in the AUDPC values of the treatments, as determined through ANOVA and Tukey's *post hoc* test ($P < 0.05$).

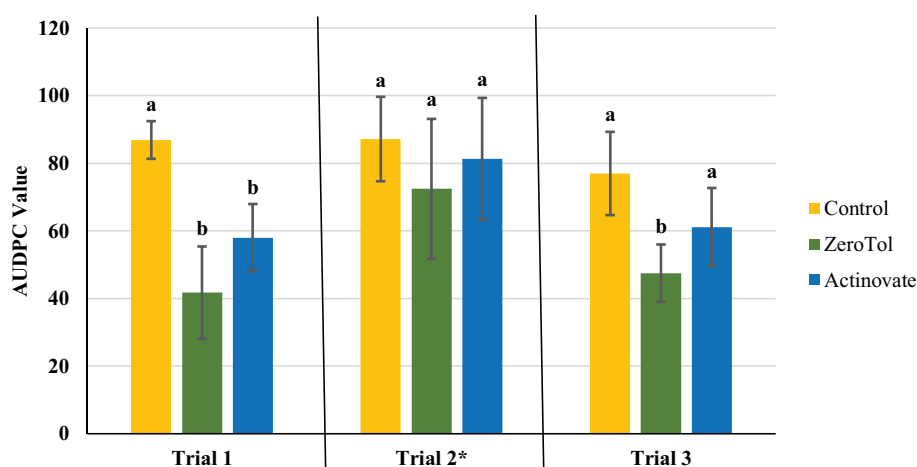


Fig. 4 Effect of four weekly applications of ZeroTol® and Actinovate® on development of powdery mildew on cannabis plants compared to a nontreated control. Data represent areas under the disease progress curves (AUDPC) from three repeated trials, each with four replicate plants (except in the case of trial 2 where an asterisk denotes three replicate plants were used instead of four). Error bars are 95% confidence intervals. Letters above the error bars represent significant differences in the AUDPC values of the treatments, as determined through ANOVA and Tukey's *post hoc* test ($P < 0.05$).

disease severity and incidence of powdery mildew, and leaves from treatments such as neem oil, Stargus, Rhapsody and ZeroTol also showed a visible reduction in disease development. By comparison, leaves treated with Actinovate and boric acid had the highest infection levels (Fig. 8).

Scanning electron microscopy

Scanning electron microscopic images showing growth of the powdery mildew pathogen on leaf surfaces of

untreated cannabis plants were compared to those following applications of disease management products. Weekly applications of three products (Rhapsody, Regalia, MilStop) or daily exposure of plants to UV-C light were made. On control leaves, mycelium from a young developing colony was observed ramifying over the surface (Fig. 9a), followed by more extensive mycelial proliferation and spore production in a mature colony on an untreated leaf (Fig. 9b). Following four applications of Rhapsody, the extent of mycelial proliferation was noticeably reduced (Fig. 9c), while some

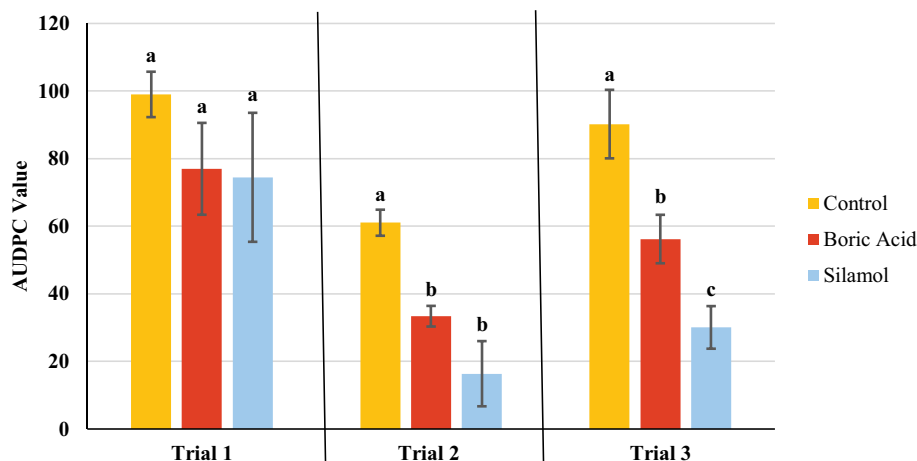


Fig. 5 Effect of four weekly applications of boric acid and Silamol[®] on development of powdery mildew on cannabis plants compared to a non-treated control. Data represent areas under the disease progress curves (AUDPC) from three repeated trials, each with four replicate plants. Error bars are 95% confidence intervals. Letters above the error bars represent significant differences in the AUDPC values of the treatments, as determined through ANOVA and Tukey's *post hoc* test ($P < 0.05$).

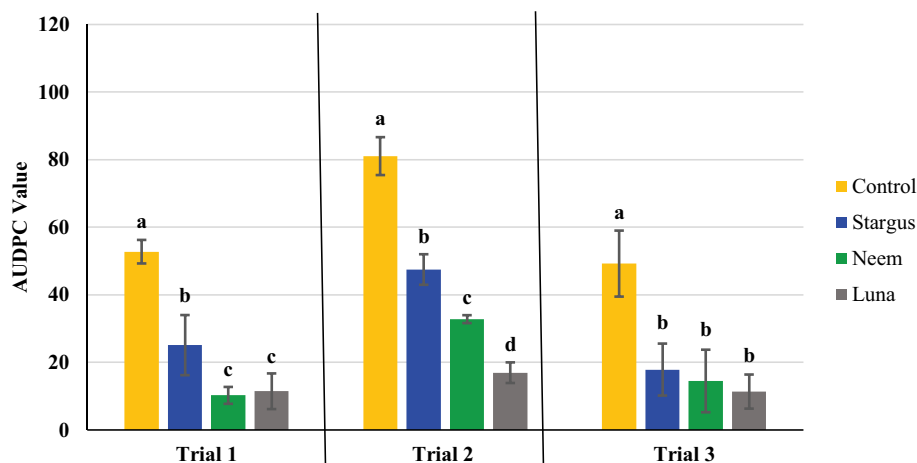


Fig. 6 Effect of four weekly applications of Stargus[®], neem oil and Luna Privilege SC 500 on development of powdery mildew on cannabis plants. Data represent areas under the disease progress curves (AUDPC) from three repeated trials, each with four replicate plants. Error bars are 95% confidence intervals. Letters above the error bars represent significant differences in the AUDPC values of the treatments, as determined through ANOVA and Tukey's *post hoc* test ($P < 0.05$).

sporulation was evident. On leaves treated with Regalia Maxx, there was an absence of mycelial growth on the leaf surface, and trichomes could be seen (Fig. 9d). Following applications of MilStop, mycelial growth was sparse and hyphae and conidia appeared to be plasmolysed (Fig. 9e). Exposure of leaves to UV light also revealed a total absence of mycelial growth (Fig. 9f). These observations show that there were obvious differences in pathogen development following various treatments when compared to the control and lend support to the disease measurement values reported above.

Effect of Regalia Maxx on chlorophyll a and b levels

Samples of leaves from plants receiving four applications of Regalia Maxx and grown in an indoor environment had mean chlorophyll *a* levels that were significantly ($P < 0.05$) higher than leaves from untreated plants, while there was no significant difference in chlorophyll *b* levels between treated and untreated plants (Fig. 10a). The chlorophyll *a/b* ratios for untreated and Regalia Maxx treated plants were 0.8 and 1.4, respectively. On plants grown in the greenhouse, there were significant ($P < 0.05$) differences between the mean levels of both chlorophyll *a* and

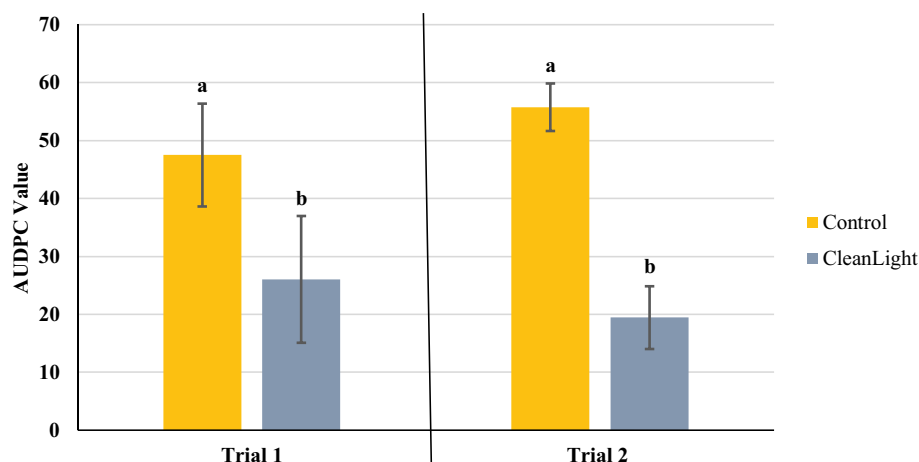


Fig. 7 Effect of daily exposure of cannabis leaves to UV-C light on powdery mildew development compared to a nontreated control. Data represent areas under the disease progress curves (AUDPC) from two repeated trials, each with eight replicate plants. Error bars are 95% confidence intervals. Letters above the error bars represent significant differences in the AUDPC values of the treatments, as determined through ANOVA and Tukey's *post hoc* test ($P < 0.05$).

chlorophyll *b* in Regalia treated leaves compared to untreated leaves (Fig. 10b). The chlorophyll *a/b* ratios for untreated and Regalia Maxx treated plants were 2.67 and 2.61, respectively. The total chlorophyll content in leaves was about three-fold higher in plants grown in the greenhouse compared with the indoor environment, likely due to differences in light intensity levels and growing conditions.

Screening of strains for susceptibility to powdery mildew

The disease scores for 12 cannabis strains are shown in Fig. 11a. Seven of the strains had significantly lower ($P < 0.05$) disease scores compared to the five highly susceptible strains, suggesting they were resistant to infection. A comparison of disease development on leaves selected from strains 'Space Queen', 'Pennywise', and 'Sweet Durga Mata' is shown in Fig. 11b.

Discussion

Powdery mildew occurs on cannabis (marijuana) and hemp plants grown indoors, in greenhouses and in outdoor environments and is a difficult disease to manage because of the rapid rate at which the pathogen can spread and a lack of available fungicides that are used to manage powdery mildew diseases on other crops (Agrios 2005). The spread of inoculum within a cannabis production facility from diseased plants is from the release of air-borne spores that are produced in large numbers (Punja 2018) as well as by infected plant materials that growers may move within and between different production facilities and from which

diseased cuttings may be taken. In this study, introduction of a diseased plant within the growing room where experiments were conducted resulted in sufficient spread and consistent disease development on control plants in repeated trials over time.

The taxonomic placement of the cannabis powdery mildew pathogen was confirmed to be in the genus *Golovinomyces* (formerly *Erysiphe*) by Pépin et al. (2018). In that study, and in subsequent work by other researchers from the United States, *Golovinomyces* has been shown to be the cause of powdery mildew on hemp (Schoener and Wang 2018; Gwinn et al. 2019; Ocamb and Pscheidt 2019; Szarka et al. 2019; Weldon et al. 2020). In these published studies, the ITS1-5.8S-ITS2 region of rDNA was used to identify the pathogen. Based on the homology to sequences previously deposited in GenBank®, three species with 100% sequence homology to each other, have been proposed to infect cannabis and hemp plants: *G. ambrosiae*, *G. spadiceus*, and *G. cichoracearum*. Further distinction among these species will require additional molecular markers other than ITS and phylogenetic analysis.

Currently, there are several products registered for use on cannabis in Canada to manage powdery mildew. These include wettable and vaporized sulphur, MilStop (potassium bicarbonate), ZeroTol (hydrogen peroxide), Regalia Maxx (an extract of giant knotweed), and Lacto-San (lactic acid) (Health Canada 2019.) Actinovate was also previously registered but has since been deregistered. Comparative efficacy studies utilizing these products for powdery mildew control on cannabis have not been previously conducted. Our



Fig. 8 A representation of the comparative efficacy of 11 treatments at managing powdery mildew development on leaves of cannabis strain ‘Copenhagen Kush’. (a) Untreated control. (b) CleanLight Pro Unit. (c) Luna SC Privilege 500. (d) Regalia Maxx. (e) MilStop. (f) Neem oil. (g) Stargus. (h) Rhapsody ASO. (i) ZeroTol. (j) Silamol. (k) Boric acid. (l) Actinovate. Representative leaves were sampled from plants in each of the treatments and photographed to show the differences in product efficacy.

results confirm that MilStop and Regalia Maxx were the most effective at reducing powdery mildew infection and spread when applied weekly. Biocontrol products

such as Rhapsody ASO and Stargus (both containing *Bacillus* sp.) also suppressed powdery mildew to some degree. Lower efficacy was observed with other

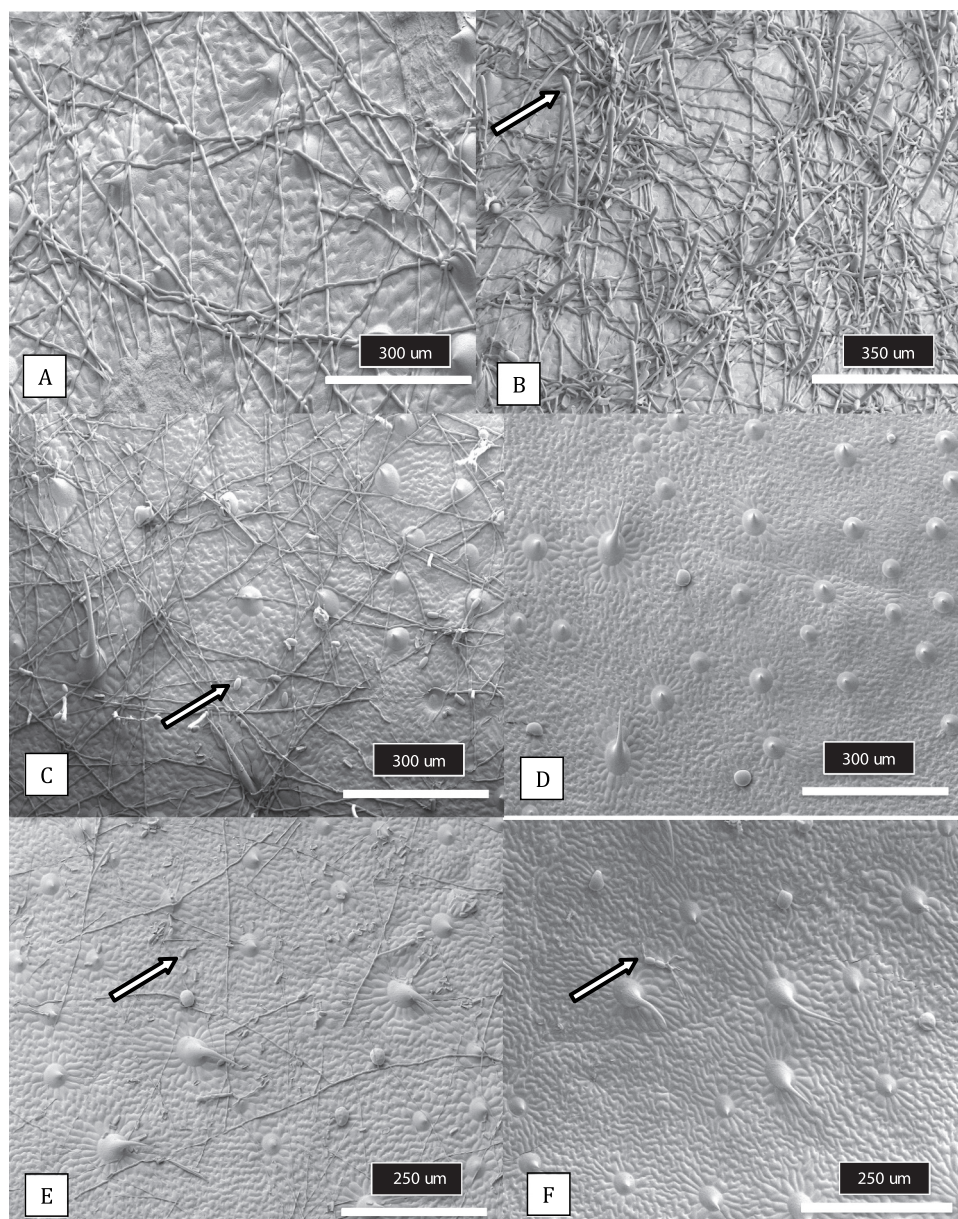


Fig. 9 Scanning electron microscopic images showing growth of the powdery mildew pathogen on leaf surfaces of untreated cannabis plants or following applications of disease management products. Weekly applications of three products or daily exposure of plants to UV-C light were made as indicated. Samples were collected after four applications of products or following a 28-day exposure to UV-C light. (a) Mycelium of a young developing colony on an untreated leaf. (b) Mycelium and spore production (arrow) of a mature colony on an untreated leaf. (c) Reduced mycelial growth on the leaf surface, with some sporulation (arrow) following applications of Rhapsody. (d) Total absence of mycelial growth on the leaf surface following applications of Regalia[®]. Hair-like projections that are visible are the trichomes. (e) Sparse mycelial growth and collapsed conidia (arrow) following applications of MilStop[®]. (f) Total absence of mycelial growth following daily exposure to UV-C light. Arrow shows conidia attempting to germinate.

products tested, such as boric acid, Silamol, ZeroTol, and Actinovate. While the fungicide Luna Privilege SC 500 was the most effective product in reducing disease development, it is not registered for use on cannabis in

Canada or in the United States and was included in this study for comparative purposes. In a recent study, Betz and Punja (2020) demonstrated that Regalia Maxx effectively controlled powdery mildew development

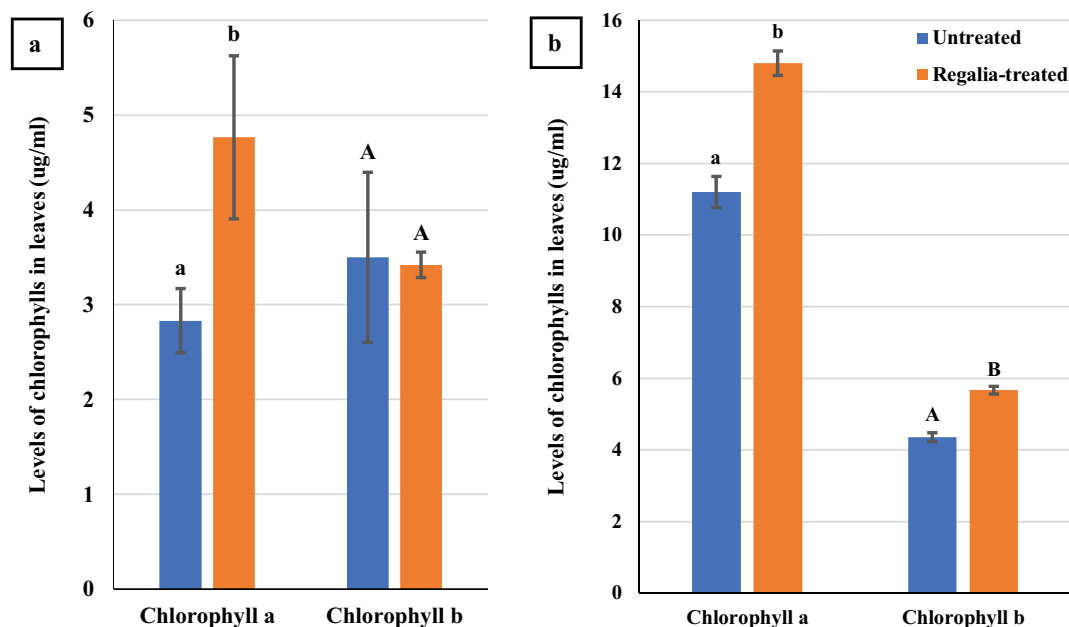


Fig. 10 Effect of four weekly applications of Regalia Maxx on mean chlorophyll *a* and *b* levels in leaves of: (a) indoor grown plants ($n = 3$) and (b) greenhouse grown plants ($n = 5$) compared to their respective controls. Levels of chlorophyll *a* were significantly increased in both trials, while chlorophyll *b* was increased in the greenhouse trial. Error bars are 95% confidence intervals. Letters above the error bars represent significant differences in chlorophyll levels, as determined through ANOVA and Tukey's *post hoc* test ($P < 0.05$).

on wasabi plants (caused by *Erysiphe cruciferarum*), while Rhapsody also reduced disease progression to a lesser extent and Actinovate was not effective. On cucumber plants, Ni and Punja (2020b) demonstrated that weekly applications of Rhapsody reduced powdery mildew development (caused by *Podosphaeria xanthii*) to the same extent as the fungicide Luna Privilege SC 500.

The active compound in Luna Privilege SC 500 (flupyradimor) is a systemic fungicide that has been shown to control both powdery mildew and leaf spot on tart cherry (Proffer et al. 2012), with preventative and moderately curative effects on grey mould caused by *Botrytis cinerea* on strawberry (Veloukas and Karaoglanidis 2012) and on table grapes (Vitale et al. 2016). One application of Luna Privilege SC 500 significantly reduced powdery mildew development on treated cannabis plants in all three trials in this study. It could be an effective treatment to manage powdery mildew on stock (mother) plants to prevent subsequent spread of the pathogen during vegetative propagation. Product registration for use on cannabis is still required in Canada.

Regalia Maxx is a formulated extract of giant knotweed (*Reynoutria sachaliensis*) that has been shown to induce systemic resistance in plants, resulting in the production and accumulation of phytoalexins, phenolic compounds

(Daayf et al. 1997, 2000) and flavonoids (Fofana et al. 2002; McNally et al. 2003). Increased activity of pathogenesis-related proteins such as glucanase, chitinase, and peroxidase has also been reported (Schneider and Ullrich 1994). Regalia was previously shown to be effective in reducing powdery mildew development on a number of crops, including cucumber (Fofana et al. 2002; Rur et al. 2018), tomato (Konstantinidou-Doltsinis et al. 2006), squash (Zhang et al. 2016), and wasabi (Betz and Punja 2020). When tested on cannabis, Regalia Maxx reduced powdery mildew development in all three trials, showing efficacy under a range of disease pressures. The mechanism of action may include production of the previously mentioned biochemical compounds, but this requires further study. Scanning electron microscopic observations showed that Regalia-treated leaves had no visible development of mycelium on the leaf surface. This may be the result of a combination of fungitoxic and induced resistance mechanisms.

It was observed that applications of Regalia Maxx caused a dark green appearance on leaves and significantly increased chlorophyll *a* levels were observed although untreated and Regalia Maxx treated plants had similar chlorophyll *a/b* ratios. The enhanced chlorophyll levels may provide a benefit to the plants, as powdery mildew infection is known to decrease photosynthesis

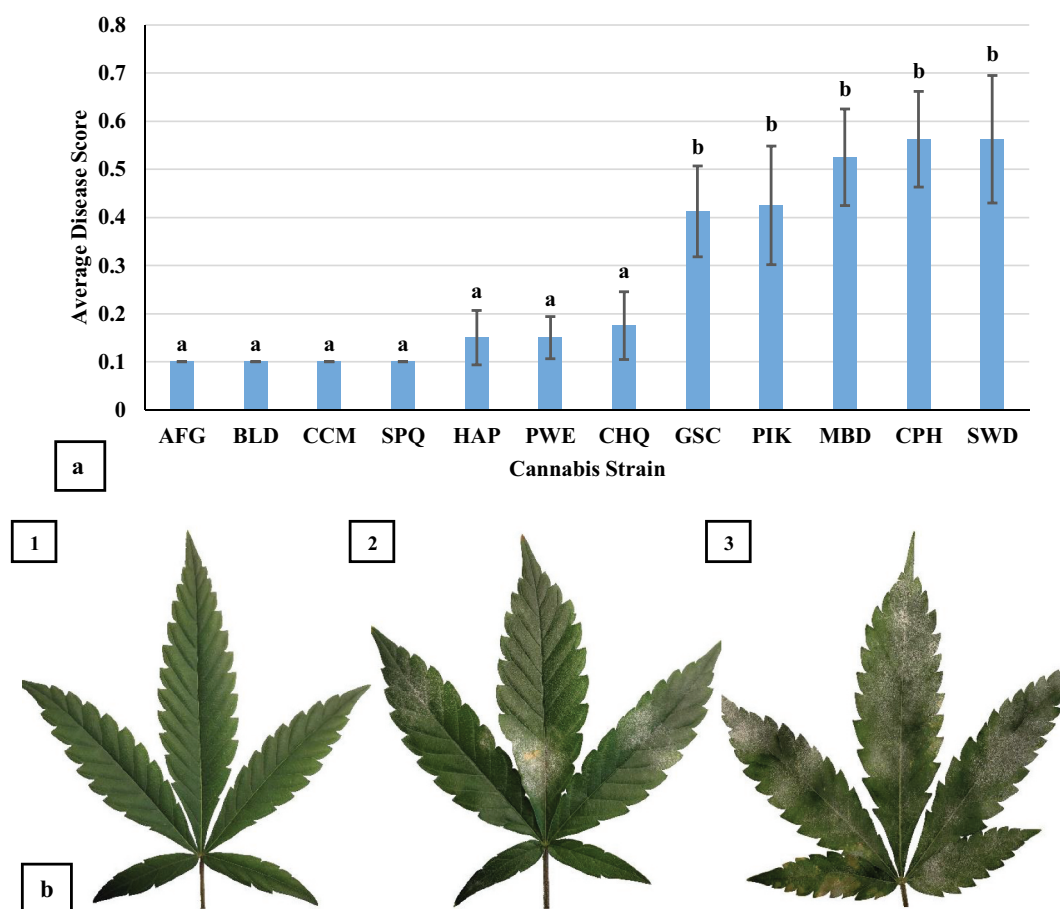


Fig. 11 (a) Response of 12 different strains (genotypes) of cannabis plants to powdery mildew infection. Mean disease score values were calculated for infected plants of each strain to evaluate their susceptibility to powdery mildew two weeks after pathogen introduction. Error bars are 95% confidence intervals. Letters above the error bars represent significant differences in the AUDPC values of the treatments, as determined through ANOVA and Tukey's *post hoc* test ($P < 0.05$). The strain abbreviations are: AFG = 'Afghani'; BLD = 'Blue Deity'; CCM = 'Critical Cali Mist'; SPQ = 'Space Queen'; HAP = 'Hash Plant'; PWE = 'Pennywise'; CHQ = 'Cheese Quake'; GSC = 'Girl Scout Cookies'; PIK = 'Pink Kush'; MBD = 'Moby Dick'; CPH = 'Copenhagen Kush'; and SWD = 'Sweet Durga Mata'. **(11B)** Comparison of powdery mildew development on leaves of three strains of cannabis. Representative leaves were sampled to illustrate a range of susceptibility to disease. (1) 'Space Queen'. (2) 'Pennywise'. (3) 'Sweet Durga Mata'.

levels in infected leaf tissues (Gordon and Duniway 1982; Balkema-Boomstra and Mastebroek 1995).

Formulated potassium bicarbonate products, including MilStop and Armicarb[®], act as fungitoxic compounds through a range of mechanisms, including altering the pH of leaf and fruit surfaces and osmotic pressure, which in turn inhibit the development of fungal mycelium and spores (Mitre et al. 2018). Potassium bicarbonate products have been reported to be effective at controlling apple scab and powdery mildew on apples (Jamar et al. 2008; Mitre et al. 2018) and significantly reduced the incidence and severity of powdery mildew on gooseberry (Wenneker 2016). Moyer and Peres (2008) demonstrated that when applied to gerbera daisies to control powdery mildew,

MilStop was effective at reducing disease, and depending on the cultivar of daisy, it was comparable to treatments of systemic fungicides. Scanning electron microscopic observations of MilStop treated cannabis leaves showed significantly reduced mycelial development compared to the control, with some evidence of plasmolysed cells that may be the result of osmotic or pH changes.

Two *Bacillus* spp. products were tested to evaluate their efficacy at managing powdery mildew on cannabis plants. These were Rhapsody ASO, which contains *Bacillus subtilis* strain QST 713, and Stargus, which contains *Bacillus amyloliquefaciens* strain 727. *Bacillus* spp. are known to produce many antifungal compounds, including the lipopeptides iturins and fengycins (Romero et al. 2007;

Siahmoshteh et al. 2018; Ni and Punja 2019), proteases (Takami et al. 1989; Degering et al. 2010), and siderophores (Shoda 2000), which act collectively to reduce pathogen development. Additionally, application of these bacteria induces host resistance that can suppress subsequent infection by pathogens (Ongena et al. 2005; Lahlali et al. 2013) as well as enhance plant growth (Kloepper and Ryu C-M 2004; Pérez-García et al. 2011). *Bacillus* based products have also been shown to be effective in managing powdery mildew, particularly when plants are treated preventatively, or if it is used in combination with systemic fungicides (Keinath and DuBose 2004; Romero et al. 2007; Gilardi et al. 2008; Elmhirst et al. 2011; Ni and Punja 2019). Scanning electron microscopic observations of treated cannabis leaves showed that Rhapsody applications reduced mycelial growth to some extent, but visible development of the pathogen still occurred and spores were produced.

Neem oil from the evergreen tree *Azadirachta indica* is known to contain an array of antifungal compounds, with the triterpenoids present acting in a synergistic or additive manner in order to achieve their antifungal effects (Govindachari et al. 1998). When neem oil fractions were purified and single triterpenoids were tested against a range of common plant pathogens, such as *Fusarium oxysporum* and *Alternaria tenuis*, there was a loss of effect (Govindachari et al. 1998). When applied to whole plants, neem products had a significant effect on reducing powdery mildew disease development on rose (Pasini et al. 1997), pea (Patil et al. 2017; Vivek et al. 2017; Deshmukh et al. 2018), okra (Moharam and Obiadalla Ali 2012), and other crops. Our results are consistent with previously published reports that show that neem oil or neem extracts are effective at reducing powdery mildew development. There are several azadirachtin or neem oil extract-based products registered for use on cannabis and hemp in the USA, but no products with these active ingredients are registered for use on cannabis in Canada.

The use of ultraviolet light (UV-C and UV-B) to reduce pathogen and mould development in agricultural production has been previously described, including for management of post-harvest decay of strawberries, potatoes, and tomatoes (Liu et al. 1993; Stevens et al. 1999; Nigro et al. 2000). As well, UV light has been used for management of powdery mildew on crops such as roses, strawberries, cucumbers, and tomatoes (Willoquet et al. 1996; Suthaparan et al. 2012, 2014, 2016) and as a seed treatment to reduce fungal infection (Siddiqui et al. 2011). The CleanLight Pro unit, which uses UV-C light, was effective at reducing powdery

mildew development in both trials conducted, with few powdery mildew colonies developing once treatment was started. Previous studies suggest that the efficacy of UV light in managing diseases is due to both its direct germicidal activity as well as its indirect ability to induce defence responses in plants, resulting in increased levels of phenolic compounds and pathogenesis-related proteins (Douillet-Breuil et al. 1999; Bonomelli et al. 2004; Hijnen et al. 2006). Scanning electron microscopic observations of treated cannabis leaves showed that mildew spores had not germinated, and mycelial growth was inhibited, suggesting a direct toxic effect. The application of UV-C light could be done on a larger scale in current cannabis production facilities through the use of mounted booms or in-row units which are available commercially and utilized for other crops (Gadoury 2019).

The use of boron to manage diseases as a soil amendment, as well as a foliar application, has been previously investigated. Boron applied to soil was reported to significantly reduce the development of clubroot on canola in field trials (Deora et al. 2011). However, when applied as a foliar treatment in the form of boric acid to field grown potatoes, it did not have a significant effect on late blight and was ineffective against *Phytophthora infestans* growth based on ED₅₀ data (Frenkel et al. 2010). Boric acid as a foliar treatment decreased the severity of late blight in potted greenhouse tomatoes, both on leaves that were treated and in distal leaves, possibly through the induction of phenolic compounds and peroxidase enzymes (Frenkel et al. 2010). Boric acid as a foliar spray also significantly reduced the number of lesions per flag leaf when used to treat tan spot on wheat (Simoglou and Dordas 2006). Canola plants treated with boric acid had twice the levels of boron in leaves compared to control plants and development of *Sclerotinia sclerotiorum* was significantly reduced (Ni and Punja 2020a). In our study, although applications of boric acid provided a significant reduction in disease in two out of three trials, the low average disease reduction in all three trials (less than 50%) indicates it is not an effective option for powdery mildew management.

Silicon is known to be effective at managing powdery mildew when applied as an amendment to soil or in nutrient solution as well as a foliar spray (Bowen et al. 1992; Schuerger and Hammer 2003; M-H et al. 2007; Shetty et al. 2011). When applied to the roots, soluble silicon products resulted in increased silica deposition in leaves, especially in the apoplast of epidermal cells as well as induction of host defences. This host response included silica deposition around areas of appressoria penetration, resulting in a reduction in haustoria formation, and an

increase in the production of phenolic compounds (Menzies et al. 1991; Samuels et al. 1991; Shetty et al. 2011). When applied as a foliar treatment, plants accumulated considerably lower levels of silicon, suggesting that management of disease from foliar silica products is due to a direct interaction with the pathogen rather than through a plant mediated mode of action (M-H et al. 2007). It is possible that the efficacy of Silamol is partially due to the low pH of the solution, with the rate used in our trials having a pH of 3.2. In two of three trials, Silamol significantly reduced powdery mildew development on cannabis plants when applied as a foliar spray. Silamol was not effective in the first trial due to the higher initial disease levels, suggesting that it is most effective when used as a preventative treatment under low disease pressure. The efficacy of silicon applied as a fertilizer to the roots of cannabis plants on powdery mildew suppression has not been investigated.

There is a limited amount of research on the efficacy of hydrogen peroxide, present in products such as ZeroTol, for disease management. Hydrogen peroxide has been shown to have some efficacy at reducing powdery mildew development on cucumber plants (Hafez et al. 2008) and for post-harvest reduction of *Botrytis cinerea* infection on white pepper fruits (Hafez 2010) as well as inducing resistance to downy mildew in pearl millet (Geetha and Shetty 2002). Hydrogen peroxide may work through two modes of action – it is known to have biocidal effects and may act as an inducer to trigger the hypersensitive response and signal transduction, inducing genes involved in systemic acquired resistance (Allan and Fluhr 1997; Alvarez et al. 1998; Kuźniak and Urbanek 2000; Hückelhoven and Kogel 2003). Our results showed that ZeroTol was partially effective in reducing powdery mildew development on cannabis plants, but was not as effective as some of the other treatments described above.

Previous research has shown Actinovate AG to be effective at reducing powdery mildew disease severity compared to unsprayed controls but it was ineffective compared to a water control on grape (Janousek et al. 2009), and compared to both water and unsprayed treatments, it was ineffective at lowering disease incidence. On squash and cantaloupe, Actinovate alone was not found to reduce powdery mildew severity compared to a water control (Zhang et al. 2011). The efficacy of Actinovate is most likely due to antibiotic and chitinase production by *Streptomyces* spp. as well as induction of plant defences following application (Conn et al. 2008; Vurukonda et al. 2018). However, in this study, Actinovate did not provide

significant disease reduction and it is not currently registered for disease control on cannabis in Canada.

Genetic resistance has been one of the most effective approaches for powdery mildew management on a wide range of crops. The genetic basis of powdery mildew resistance is in part based on R-genes, known to be present in crops such as tomato, cucumber, wheat, hops, melon, and numerous others (Huang et al. 2000; Bai et al. 2003; Sakata et al. 2006; Cao et al. 2011; Ning et al. 2014; Wolfenbarger et al. 2014). The mechanistic basis for resistance of plant tissues to powdery mildew infection is complex, and numerous changes in gene expression and biochemical compounds are described (Qiu et al. 2015). Among the 12 cannabis strains that were tested for their susceptibility to powdery mildew in this study, it was observed that seven showed partial or complete resistance to infection. This qualitative rating may suggest that one major gene for resistance is present. Differing levels of susceptibility to powdery mildew infection have also been observed among hemp cultivars (Cala et al. 2019). The basis for this resistance in cannabis and hemp is not yet known. The isolation and cloning of R-genes in the *C. sativa* genome to powdery mildew resistance is needed to enhance breeding of cannabis strains with disease resistance.

This research represents an initial investigation into the management of powdery mildew on cannabis. Further research is needed to determine how potential adjustments to the rates of these products may affect their efficacy, the possible benefits to using products in tandem (alternating Regalia Maxx and MilStop rather than just using MilStop, for example), as well as the effects of spraying these products at different intervals, not just weekly. In addition, further research is needed to establish modes of action of some of the most effective products described in this study.

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